IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

Mathematics for Engineers II Part A

Tuesday 10 January 2023 10:00

Time allowed: 1:30 hours

There are TWO questions on this paper.

Answer BOTH questions.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : D. Nucinkis

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THE QUESTIONS

1. **[20]**

a) Consider the mapping

$$w = \frac{1}{(z-1)^2}$$

where $z = x + iy$ and $w = u + iv$ where $x, y, u, v \in \mathbb{R}$. Show that all circles in the z-plane with a centre on the real axis at $z = 1$ and radius R map to circles in the w-plane, centred at the origin and with radius $1/R^4$. [5]

b) i) Find the three poles of $f(z)$ and the residues at these poles for

$$f(z) = \frac{e^{iz}}{z(z^2+1)}$$

[6]

ii) Show that

$$\lim_{r \to 0} \int_{\Gamma(r)} \frac{e^{iz}}{z(z^2+1)} = i\pi$$

where Γ is an anticlockwise semicircular path of radius r in the lower half plane [Im(z)<0], centred at the origin with the integral taken at the limit where $r \to 0$.

[3]

iii) Hence by choosing a suitable contour in the complex plane, use Jordan's Lemma and the Residue Theorem to show that

$$\int_{-\infty}^{\infty} \frac{sinx\ dx}{x(x^2+1)} = \frac{\pi}{e}(e-1)$$

[6]

2. **[30]**

a) i) Explain diagrammatically how the Gram-Schmidt process produces an orthonormal set of vectors $\mathbf{q}_i$ from a non-orthogonal set of vectors $\mathbf{a}_i$ that span $\mathbb{R}^n$ from its algorithm:

$$\mathbf{x}_1 = \mathbf{a}_1;\ \mathbf{x}_i = \mathbf{a}_i - \sum_{j=1}^{i-1} \frac{\mathbf{x}_j{}^T \mathbf{a}_i}{\mathbf{x}_j{}^T \mathbf{x}_j} \mathbf{x}_j, 1 < i \leq n;\ \mathbf{q}_i = \frac{\mathbf{x}_i}{|\mathbf{x}_i|}$$

[4]

ii) Hence produce a set of orthonormal vectors from

$$\mathbf{a}_1 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \mathbf{a}_2 = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}, \mathbf{a}_3 = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

[3]

b) The coupled motion of a system can be described by the matrix equation:

$$\dot{\mathbf{x}} = A\mathbf{x}$$

where **x** is a vector of the coupled variables and A describes the coupling.

i) Show by expressing A in diagonal form, $A = S\Lambda S^{-1}$ and introducing a second variable $\mathbf{x} = S\mathbf{y}$, the matrix equation can be solved for initial conditions $\mathbf{x} = \mathbf{x}(0)$ as

$$\mathbf{x} = Se^{\Lambda t}S^{-1}\mathbf{x}(0)$$

[5]

ii) Hence solve **x** for the coupled equations:

$$\dot{x}_1 = 5x_1 + 2x_2$$
$$\dot{x}_2 = 2x_1 + 5x_2$$

when $x_1(0) = 1, x_2(0) = 0$. [5]

c) A m x n image is represented using its first p principal components in a singular value decomposition (SVD).

i) Show that the number of unique matrix elements needed to represent the three matrices of the SVD decomposition of this image when only the first p a principal component analysis (PCA) are retained is given by $p(m+n+1)$. [4]

ii) Hence derive an expression for the maximum value of p for PCA to offer a lower number of elements than the original image and that value for a 1000 x 1000 image [2]

iii) The variance of the contribution of the nth principal component to the total variance of a matrix is given by σ_n^2. It is found that the principal components of most images are approximately related to their rank n by $\sigma_n \sim 1/n$. Show on that basis that approximately $\frac{197}{24\pi^2}$ ~83% of the variance of a large image is captured by the first four principal components, noting that

$$\lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{r^2} = \frac{\pi^2}{6}$$

[7]